Resource Considerations for Establishing and Operating a Federated Human Data Node

Mallory Ann Freeberg[1], Salvador Capella-Gutierrez[2], Amy Curwin[3], Thomas Keane[1], Giselle Kerry[4], Krzysztof Kochel[5], Melissa Konopko[4], Błażej Marciniak[5], Kjell Petersen[6], Jordi Rambla[3], Dylan Spalding[8], Johan Viklund[7]

1 EMBL European Bioinformatics Institute; 2 Barcelona Supercomputing Center; 3 Centre de Regulació Genòmica; 4 ELIXIR Hub; 6 University of Bergen, ELIXIR Norway; 7 Uppsala University, ELIXIR Sweden; 5 Biobank Laboratory, Department of Molecular Biophysics, Faculty of Biology and Environmental Protection, University of Lodz; 8 CSC-IT Center for Science, ELIXIR Finland

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Abstract

Most individual-level human omics data to-date have been generated in the context of biomedical research projects and are shared via global repositories such as the European Genome-phenome Archive (EGA). Many countries now have emerging personalised medicine programmes funded by national or regional healthcare initiatives. Data generated in healthcare contexts are subject to more stringent governance than research data. To address this need, the Federated EGA (FEGA) provides a network of connected resources to enable transnational discovery of and access to human data while respecting jurisdictional data protection regulations. FEGA nodes represent diverse use-cases ranging from national platforms managing citizen's omics data to community-based healthcare repositories. Key to developing a FEGA node is determining budget and resource requirements. These guidelines will support human resource, services, infrastructure, and their impact on financial planning for ELIXIR nodes interested in joining the Federated EGA network of human data resources.

Resourcing a Federated Human Data Node: Lessons from Federated EGA

Federated European Genome-phenome Archive

Governance

Establish governance structure

- Establish Service Provider legal entity
- Acquire funding, resource prioritisation

"ELIXIR Norway is a consortium and mandated University of Oslo to provide the FEGA Norway node on behalf of ELIXIR Norway."- NO

"The Polish node is founded with UE grant; storage is provided by governmental cloud." - PL

Develop vision & strategy

- Approve strategic roadmap
- Identify funding for immediate and long-term resources

"Aligning development and strategic aims with other infrastructures and projects, as well as national use cases, are necessary for long-term sustainability and to maximise interoperability and efficiency." - FI

Infrastructure

Hardware requirements

Storage capacity & robustness

Network capacity & security

Compute capacity

"Our initial deployment is modest, but needs to be set up over two security zones that adds complexity." - NO

"Implementation is based on premise Kubernetes clusters. Storage is provided by the government vendor." - PL

Software requirements

Develop, maintain, and deploy local node software to support data submission, archive, discovery, access, and analysis.

"Collaboration with neighbouring ELIXIR nodes in the Nordics has facilitated an above critical mass development team focused on FEGA microservices" - SE, NO, FI

"Strong synergies with CEGA allows cross-validation of technological developments." - ES

Legal

Develop & approve node data protection & security policies

"We worked together with Security Officer, Data Protection Officer, and Data Privacy experts of UiO for a 1,5-2 year period." - NO

Develop & approve node legal agreements with users

- Data Processing Agreement
- Data Access Agreement

"Getting agreement templates approved by institutional legal teams took longer than we anticipated. Had we known, we would have started the approval process earlier." - CEGA

Approve FEGA Legal Agreement

"It took three iterations of the agreement & one year of cooperation and consultations with DPO, Legal, IT Department to approve." - PL

Operations

Train internal node staff

"It's critical to build early a joint team understanding across borders between developers, admins, helpdesk, and leaders." - NO

Example minimum personnel to establish and operate a node:

Establish → 4-6 people with 2-3 FTE

Operate → 6-8 people with 4-5 FTE

Establish node user support systems

Helpdesk

SOPs

Training materials

"We operate a Helpdesk with dedicated staff who handle all user queries and escalate issues to operations teams, when needed." - CEGA

Data & metadata

Establish data security practices

- Security breach and incident response plan
- Risk register and periodic assessment
- Risk and Vulnerability Analysis and periodic assessment

"ELIXIR-SE node hosting university has well worked out security guidelines and practices that we have followed." - SE

Establish data flow processes

- How to receive data in and serve data out
- Internet facing and internal orchestration

"We established tailored processes to interface the major national infrastructure for computing on sensitive data" - SE

Key: CEGA: Central EGA; **FI**: FEGA Finland; **NO**: FEGA Norway; **PL**: FEGA Poland; **ES**: FEGA Spain; **SE**: FEGA Sweden

Community & outreach

Engage with community

Build **trust** with potential data submitters

"We have targeted large University hospitals and National Biobank infrastructure projects for regular updates and discussions on our emerging node." - NO

"We leveraged our established network with national sequencing facilities and our bioinformatics support experts to establish contact with data submitters." - SE

"We attend scientific conferences of societies that are involved in the generation and processing of human data." - PL

"We are promoting best practices on richly described genomics data among local submitters at healthcare settings." - ES

Train users & stakeholders

- Collect and assess use cases
- Organise and run hands-on workshops
- Incorporate feedback into node roadmap

Additional resources

- More about ELIXIR Federated Human Data Community
  - https://elixir-europe.org/communities/human-data
  - Contact: Giselle Kerry (giselle.kerry@elixir-europe.org)
- More about Federated EGA
  - https://ega-archive.org/federated
  - FEGA Maturity Model: https://inab.github.io/fega-mm/
  - Contact: Mallory Freeberg (mfreeberg@ebi.ac.uk) and Amy Curwin (amy.curwin@crg.eu)

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